

SUMMARY

Senior data scientist building production ML, multi-agent AI, and decision systems from problem framing through deployment and monitoring. Ph.D. researcher and open-source ML author whose software is used by 100+ research labs.

EXPERIENCE

Huntington National Bank · Senior Data Scientist

2023 – Present

- Agentic AI.** Lead a four-layer multi-agent architecture spanning specialist agents, agent-human collaboration, orchestration and reconciliation, and source-data verification; being scaled across the commercial bank for hundreds of bankers, reducing customer research from hours to minutes and expanding into additional business segments.
- Intelligent applications.** Built and productionized decision applications using entity resolution, fuzzy matching, dynamic calculations, forward-looking segmentation, and executive outreach planning; support decision-making across thousands of commercial relationships and prospects through automated, scheduled production delivery.
- Production ML & reliability.** Built and productionized recommendation models generating thousands of refreshed deepening recommendations monthly across 30+ banking products, with multiple-fold lift over baseline targeting. Automated production workflows that previously required 1-2 days of manual work per cycle, with monitoring across data quality, model performance, and downstream decision usefulness.
- Technical leadership.** Lead 4 senior data scientists across 8 concurrent AI, ML, and analytics workstreams spanning commercial growth, customer engagement, data enrichment, and decision support; translate business challenges and emerging ideas into scoped technical initiatives, priorities, and delivery plans while facilitating cross-functional planning with business, product, sales, analytics, and leadership stakeholders.

Carnegie Mellon University · Ph.D. Researcher, Neuroscience

2018 – 2023

- B-SOiD.** Created an open-source unsupervised behavior-discovery pipeline and no-code application; published in *Nature Communications* and adopted by 100+ labs / 250+ users.
- A-SOiD.** Co-first-authored an active-learning platform that surpassed all 49 Alcrowd challenge entries using 12% of the training data; published in *Nature Methods*.
- Neural decoding.** Built a 24/7 recording system and cloud pipeline processing 160 TB of synchronized video and neural data to decode behavior directly from neural activity.

PERSONAL PROJECT

Vowel for Weddings · Full-Stack iOS App

2026

Designed, built, and shipped a native SwiftUI app for my wedding, used by 30+ guests for event information, photo discovery, and guest assistance; built with a Supabase backend, grounded LLM assistant, face matching, and push notifications.

SKILLS

Machine learning	Classical ML (scikit-learn) · gradient boosting (LightGBM, XGBoost) · deep learning (PyTorch, TensorFlow / Keras) · learning-to-rank & recommender systems · propensity modeling · active learning · dimensionality reduction (UMAP) · clustering (HDBSCAN) · evaluation (AUC, lift)
Applied AI	Amazon Bedrock · multi-agent orchestration · RAG / grounding · human-in-the-loop systems · verification & structured evaluation
Production ML	AWS SageMaker · Snowflake · scheduled AWS pipelines · Shakudo · Evidently AI · Anomalo · model monitoring · model governance & validation (MRM)
Languages	Python · SQL · Swift · TypeScript / JavaScript
Applications & data tools	FastAPI · React · Streamlit · Power BI · Tableau · Supabase · Fly.io · Polars / Pandas · Git · Linux

EDUCATION

Ph.D., Neuroscience	Carnegie Mellon University · 2023 Presidential Fellowship · \$50K Oracle for Research grant
M.S.	Boston University School of Medicine · 2016
B.S.	University of Illinois at Urbana-Champaign · 2013

SELECTED PUBLICATIONS

Schweihoff J.F., **Hsu A.I.**, et al. "A-SOiD, an active-learning platform for expert-guided, data-efficient discovery of behavior." *Nature Methods*, 2024. Co-first author.

Hsu A.I., Yttri E.A. "B-SOiD, an open-source unsupervised algorithm for identification and fast prediction of behaviors." *Nature Communications*, 2021.